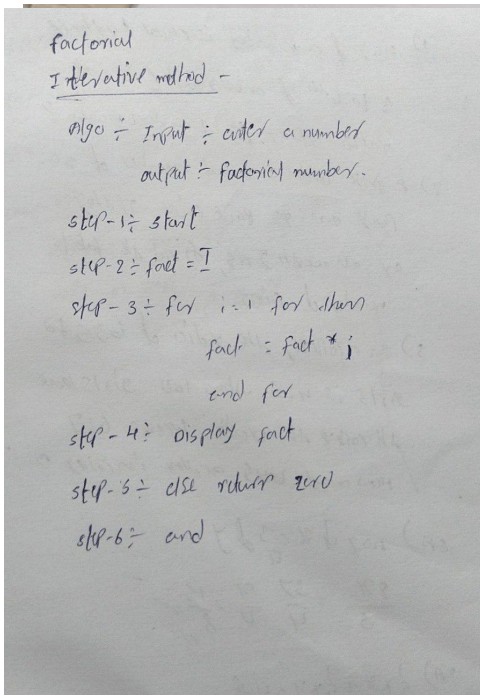
PRACTICAL-04

AIM: Implementation and Time analysis of factorial program using iterative and recursive method

1.) ITERATIVE METHOD

ALGORITHM:



PROGRAM:

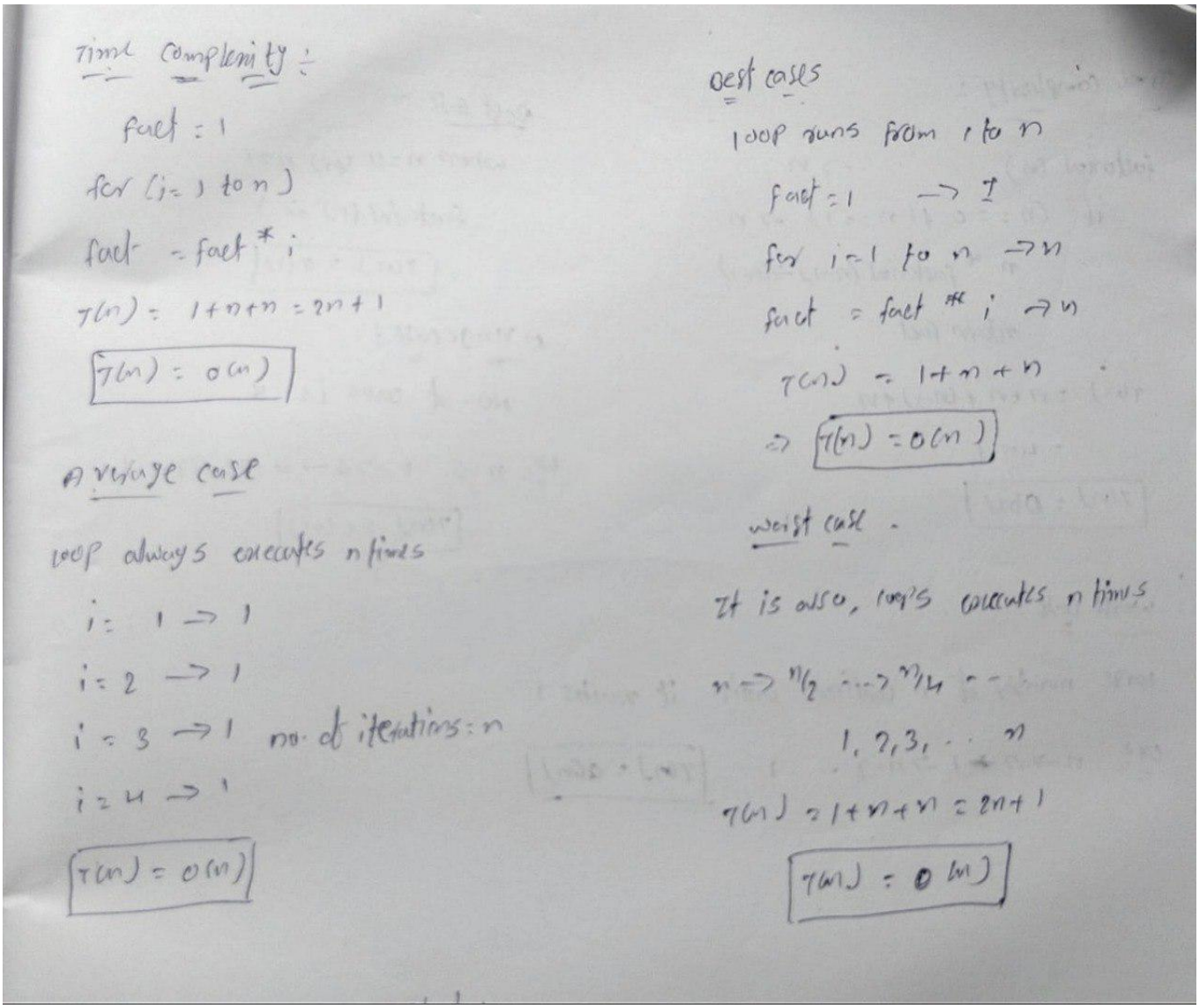
```
#include <iostream>

using namespace std;

int main(){

   int n;

   int fact = 1;

   cout << "enter a number:";

   cin >> n;

   for(int i = 1; i <= n; i++){

      fact = fact * i;

   }

   cout << "Factorial of " << n << " = " << fact;
```

```
    return 0;
}
```

OUTPUT:

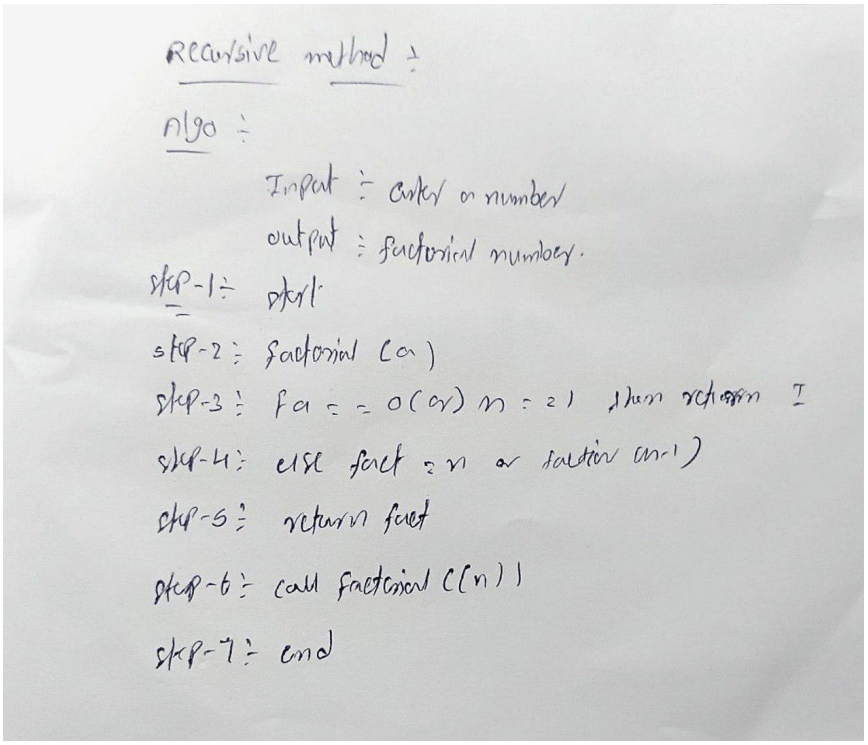
```
Output
enter a number:5
Factorial of 5 = 120

=== Code Execution Successful ===
```

TIME COMPLEXITY:



2.) RECURSIVE METHOD:

ALGORITHM:



PROGRAM:

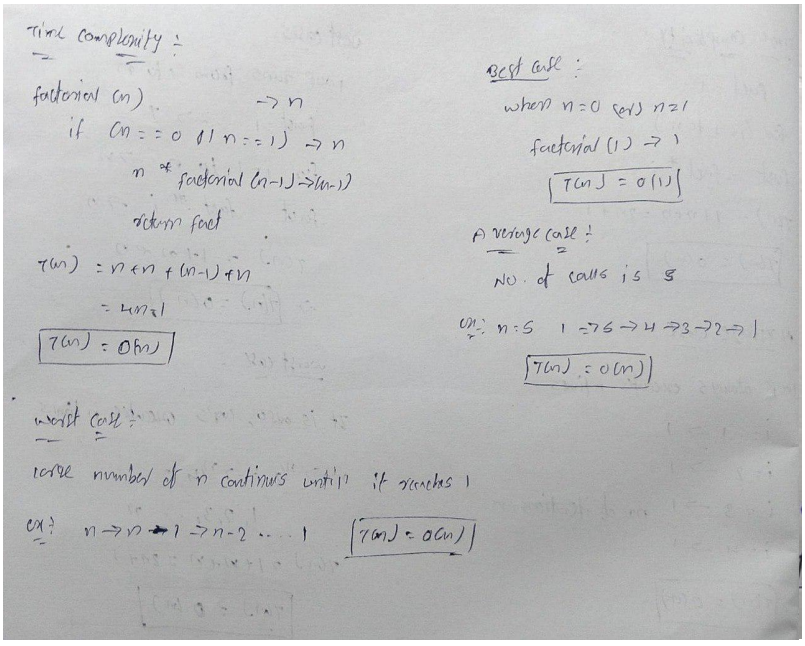
```
#include <iostream>
using namespace std;
class factorial{
   public:
   int factr(int n){
   if(n == 0 || n == 1){
      return 1;
   }
   else{
   int result = n*factr(n-1);
   return result;
   }
}
```

```
};
int main(){
    int n,result;
    factorial obj;
    cout << "enter a number:";
    cin>>n;
    result = obj.factr(n);
    cout << "factorial of " << n << " is " << result;
    return 0;
}
```

OUTPUT:

```
Output
enter a number:7
factorial of 7 is 5040

=== Code Execution Successful ===
```

TIME COMPLEXITY:



CONCLUSION:

Both iterative and recursive methods are used to find factorial. Both take O(n) time, but the iterative method uses less memory and is simpler to implement.